

Assignment 5

Ethan

Set Up

Packages

```
#general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(scales)
library(NatParksPalettes)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

Necessary Code from Class

```

# creating new object from sites
ncos_sites <- sites %>%
  # filtering to only include sites at NCOS, sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new object from taxon_list
taxon_list_clean <- taxon_list %>%
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))

water_quality_clean <- water_quality %>%
  clean_names() %>%

```

```

semi_join(ncos_sites, by = "site") %>%
unite("date_site", date, site, remove = TRUE) %>%
group_by(date_site) %>%
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) %>%
ungroup() %>%
filter(date_site != "2022-08-12NVBR")

aq_ins_clean <- aq_ins %>%
  clean_names() %>%
  semi_join(ncos_sites, by = "site") %>%
  rename(date = date_on_vial) %>%
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         annelida_polychaete,
         diptera_chironomid) %>%
  filter(sample_type %in% c("FB250", "FB 250")) %>%
  pivot_longer(
    # columns to make longer
    cols = ostracod:annelida_polychaete,
    # new column for the old column names
    names_to = "taxon",
    # new column for the values in each of the old columns
    values_to = "abundance"
  ) %>%
  group_by(date, site, taxon) %>%
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) %>%
  ungroup() %>%
  unite("date_site", date, site, remove = FALSE) %>%
  left_join(water_quality_clean, by = "date_site") %>%
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) %>%
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "East Channel",
    site == "NMC" ~ "Main Channel",

```

```

    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NDC" ~ "Devereux Creek")) %>%
mutate(site_full = fct_reorder(site_full,
                              med_sal,
                              .fun = "median",
                              .na_rm = TRUE),
       site_full = fct_rev(site_full))

copepods <- aq_ins_clean %>%
  filter(taxon == "copepod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

```

1. Visualizing differences across sites in major taxon

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: taxon
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()`

b. Wrangling the data

```

ostracods <- aq_ins_clean %>%
  filter(taxon == "ostracod") %>%
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)

```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-05-05_NEC	2022-05-05 00:00:00	NEC	ostr~	0.267	9.45	63.8	7.3
2	2023-03-03_NDC	2023-03-03 00:00:00	NDC	ostr~	5.73	7.45	0.688	5.64
3	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA

```

4 2022-05-11_NDC 2022-05-11 00:00:00 NDC ostr~ 0 9.11 0.398 9.4
5 2023-05-05_NPB 2023-05-05 00:00:00 NPB ostr~ 24.7 NA NA 12.1
6 2023-07-17_NPB 2023-07-17 00:00:00 NPB ostr~ 4.53 NA NA NA
7 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR ostr~ 4.8 7.34 4.1 6.1
8 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1 ostr~ 0.533 7.48 NA 2.38
9 2022-08-15_NMC 2022-08-15 00:00:00 NMC ostr~ 0.133 7.61 NA NA
10 2022-01-28_NDC 2022-01-28 00:00:00 NDC ostr~ 9.47 7.65 2.00 9.55
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

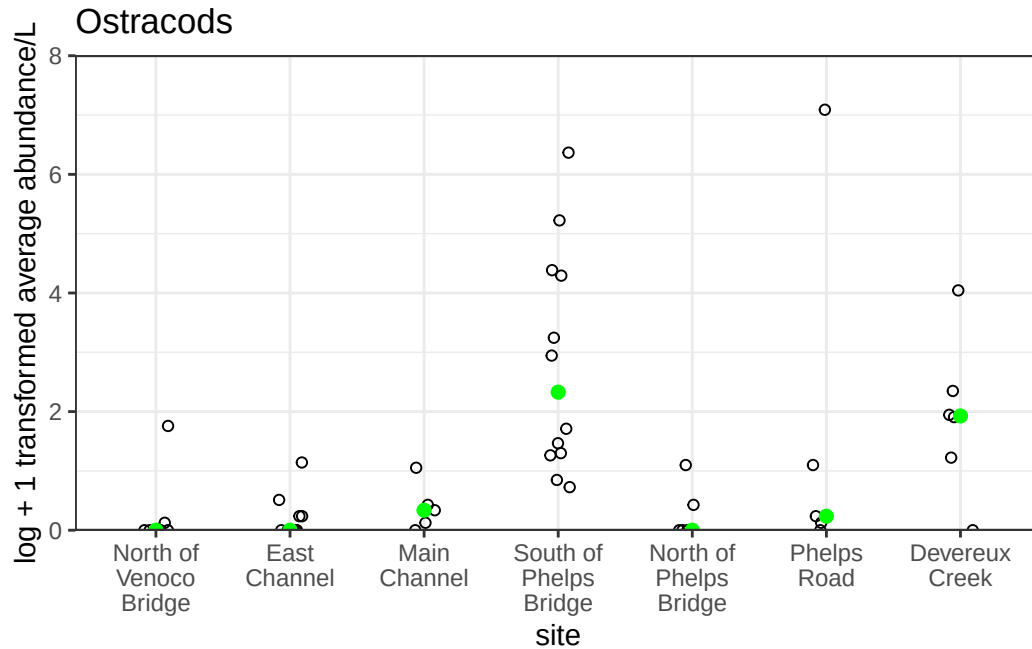
```

c. Coding the figure

```

ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit))+
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "green",
              size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 8)) +
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  theme_bw()
ostracod_plot

```



d. Creating a multipanel figure

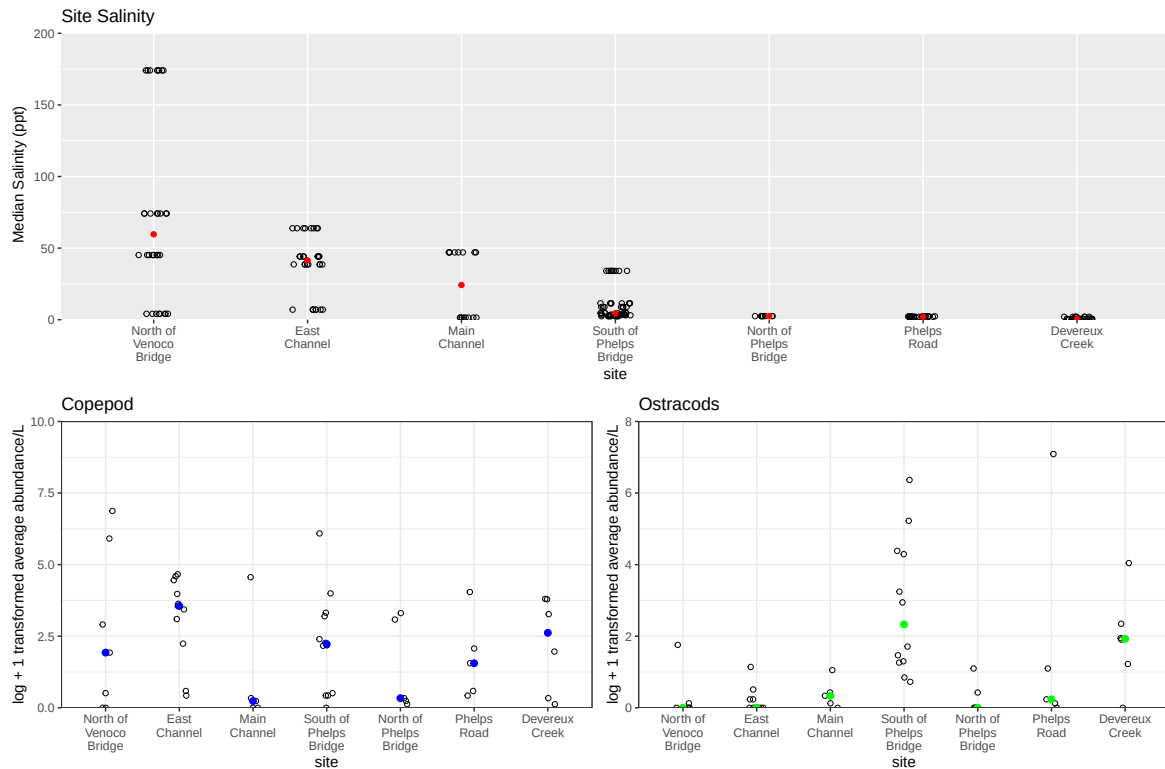
```
# creating an object of a salinity plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +

  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = 'red') +
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 200)) +
  labs(x = "site",
       y = "Median Salinity (ppt)",
       title = "Site Salinity")
```

```
# creating a plotting object of copepod abundance
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    y = log_ave_lit))+

  geom_jitter(height = 0,
             width = 0.1,
             shape = 21) +
  stat_summary(geom = "point",
             fun = median,
             color = "blue",
             size = 2)+
  scale_x_discrete(labels = label_wrap(width = 10)) +
  scale_y_continuous(expand = c(0, 0),
                    limits = c(0, 10)) +
  labs(x = "site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod") +
  theme_bw()
```

```
salinity_plot /
(copepod_plot | ostracod_plot)
```



e. Interpreting the figure

The biggest differences in copepod and ostracod abundance across sites can be seen at the East Channel site, where copepods are highest in abundance, and ostracods are extremely low in abundance. The two species are similar in abundance at the Main Channel, South of Phelps Bridge, North of Phelps Bridge, and Devereux Creek sites.

There doesn't appear to be a correlation between site salinity and copepod or ostracod abundance. Median salinity is order by descending ppt from left to right on the x-axis, and the invertebrate abundance plots with the same x-axis don't follow a descending or ascending pattern, appearing relatively random.

2. Visualizing total abundance as a function of salinity

a. Planning the Figure

- x-axis: Salinity (ppt)
- y-axis: log_avg_lit

- color: taxon
- geometry: geom_points()
- facets: facet_wrap(~ taxon)

b. Wrangling the Data

```
avg_taxon_abundance <- aq_ins_clean %>%
  select(date_site, taxon, med_sal, ave_lit) %>%
  mutate(
    taxon = to_any_case(taxon, case = "title"),
    log_ave_lit = log(ave_lit + 1),
    taxon = fct_reorder(taxon, ave_lit, .fun = "max", .na_rm = TRUE),
    taxon = fct_rev(taxon))

slice_sample(avg_taxon_abundance, n = 10)
```

A tibble: 10 x 5

	date_site	taxon	med_sal	ave_lit	log_ave_lit
	<chr>	<fct>	<dbl>	<dbl>	<dbl>
1	2021-11-23_NPB2	Ostracod	1.99	0.267	0.236
2	2023-02-12_NEC	Copepod	NA	30	3.43
3	2021-08-25_NPB	Cladocera	4.57	0	0
4	2023-02-28_NPB	Diptera Ceratopogonidae	NA	0	0
5	2021-09-01_NPB2	Copepod	NA	0.8	0.588
6	2022-08-10_NPB	Nematode	8.84	0	0
7	2021-09-01_NPB2	Annelida Polychaete	NA	0	0
8	2022-05-11_NDC	Copepod	0.398	0.133	0.125
9	2022-11-12_NEC	Hemiptera Corixidae Boatman	7.01	0	0
10	2022-02-23_NVBR	Diptera Ceratopogonidae	45.2	0.4	0.336

c. Coding the figure

```
ggplot(data = avg_taxon_abundance,
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = taxon))+
  geom_point()+
  facet_wrap(~ taxon)+
```

```

theme_bw()+
scale_color_manual(values = natparks.pals("Torres"))+
theme(legend.position = "none")+
scale_y_continuous(expand = c(0,0),
                    limits = c(-0.5,8))+
labs(x = "Median Salinity",
     y = "Log of Average Invertebrate Abundance\nper Liter + 1",
     title = "Invertebrate Abundance in Relation to Median Water Salinity")

```

